

Harmony in Motion: A Music-Integrated Therapeutic Exercise Framework Leveraging Acoustic-Inference Neural Beat Synchronization

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Abstract

There are millions of people with disorders, and still yet almost everyone deals with stress issues from time to time. In these cases, long term exercises such as physical exercise or therapies such as music therapy and psychotherapy are beneficial but are tedious and boring. Other solutions such as Virtual Reality simulation, Motor Imagery-driven Hand Redirection are effective but not all can have access to such technologies due to being highly expensive, as well as are bulky, difficult to set up and use. Video games can serve as effective tools for encouraging and helping them relax and exercise cognitively and physically. In this work, to help those with hand stiffness/motor disorder, eye coordination difficulties, reaction times, and stress issues to exercise, focus and relieve stress in a fun way, a video game involving hand movement, eye coordination and feel of playing as well as listening to music, is proposed. The game is made using unity game engine with the help of C# as the coding language. The gameplay includes repeated, timely tapping or pressing of several tiles during music beats with different difficulty modes and songs. It uses the Acoustic-Inference Neural Beat Synchronization algorithm which, in performance testing, achieved a high level of synchronization with an actual latency for the Audio Engine of 4ms and Beat Mapping of 8ms, both well within the <10ms and <15ms targets respectively. User evaluation showed a 4.2/5 score for stress reduction and a 4.1/5 for immersion, representing a +5.0% and +5.1% improvement over conventional methods.

Keywords: Unity Game Engine, C#, Acoustic-Inference Neural Beat Synchronization (AINBS), Motor Disorder Rehabilitation, Eye-Hand Coordination, Stress Relief, Beat Mapping, Real-time Audio Processing.

1. Introduction

A large number of individuals, in today's fast-paced world, suffer from various physical, motor, and psychological disorders, while many others frequently experience stress and mental fatigue [3,4,5,6,9,10,11,12]. Conditions such as hand stiffness, impaired motor control, poor eye-hand coordination, delayed reaction time, and elevated stress levels can significantly affect an individual's quality of life [3]. Conventional approaches to address these issues, including physical exercise, physiotherapy, music therapy, or psychotherapy, have proven benefits when practiced consistently over a long period [4,11]. However, these methods often become repetitive, tedious, and unengaging, leading to reduced motivation and poor adherence [6, 9, 10, 11].

Advanced technological solutions such as virtual reality simulations and motor imagery-driven hand redirection systems have also demonstrated effectiveness in rehabilitation and skill development [3,9]. Despite their potential, access to these technologies remains limited, as they are often expensive, bulky, and complex to set up and use, making them impractical for widespread or everyday application [6].

Video games have emerged as a promising alternative tool beyond entertainment. Game-based applications provide interactive, engaging, and motivating environments that encourage users to perform repetitive physical and cognitive tasks without experiencing boredom [5]. In addition to their effectiveness, video games are highly accessible, as many are free or low-cost, require little to no setup, and are easy to learn and use. Furthermore, they are widely available on smartphones, devices that most individuals already own and have regular access to, making

game-based solutions practical for daily use across diverse populations. By integrating elements such as music, visual feedback, and timed interactions, games can effectively promote motor skill development, eye–hand coordination, reaction time improvement, and mental relaxation [7].

This research work proposes the development of a music-based video game designed to assist individuals with motor and coordination difficulties, as well as those seeking stress relief. The game focuses on repeated and timely tapping or pressing of tiles synchronized with musical beats, encouraging hand movement, concentration, and rhythm-based coordination. Developed using the Unity game engine and programmed in C#, the game offers multiple difficulty levels and songs to cater to users with varying abilities and progress levels. Through this approach, the research work aims to provide an enjoyable, accessible, and effective platform for physical exercise, cognitive focus, and stress reduction.

2. Literature Survey

The landscape of modern rehabilitation is undergoing a significant paradigm shift, transitioning from clinical, therapist-led interventions to self-managed, tech-centric approaches. Given the conditions of modern age, millions of individuals struggle with physical, motor, and psychological disorders, while a vast majority of the general population faces chronic stress and mental fatigue [3, 4, 5, 6, 9, 10, 11, 12]. Conditions such as hand stiffness, impaired motor control, and delayed reaction times significantly diminish an individual's quality of life [7]. While conventional methodologies including physiotherapy, music therapy, and psychotherapy, have well-documented benefits, they often suffer from a critical flaw: low patient adherence due to the repetitive, tedious, and unengaging nature of long-term exercises [6, 9, 10, 11].

To address these motivational barriers, researchers have turned to Serious Games, applications that utilize the interactive and engaging elements of video games for purposes beyond pure entertainment [5]. By providing immersive and motivating environments, these games encourage users to perform repetitive physical and cognitive tasks without the onset of boredom. Unlike advanced technological solutions like Virtual Reality (VR) simulations or motor imagery-driven redirection systems, which are often criticized for being expensive, bulky, and difficult to set up mobile and PC-based games offer unparalleled accessibility. Given that smartphones are nearly ubiquitous, game-based solutions are now a practical tool for daily use across diverse populations.

2.1 Mechanisms of Rhythmic and Visual Intervention

The efficacy of these digital interventions relies on the integration of specific sensory stimuli:

Auditory Stimulation: Synchronizing physical movement with musical beats promotes rhythm-based coordination and cognitive focus.

Visual Feedback: Real-time visual cues (such as VFX and tile spawning) assist in eye-hand coordination and reaction time improvement.

Interactive Engagement: Game engines like Unity, combined with logic-heavy C# programming, allow for the creation of modular systems that can adjust difficulty levels in real-time to suit varying progress levels.

2.2 Objective

This literature survey evaluates ten pivotal research contributions that explore the intersection of motion tracking, neural synchronization, and gamification. These studies range from high-end VR-based motor imagery to accessible mobile platforms for elderly cognitive health. By analysing these existing solutions, this research work identifies the technical requirements for the Acoustic-Inference Neural Beat Synchronization (AINBS) algorithm, ensuring a high-performance system that maintains responsiveness and accuracy while providing significant stress relief and motor training.

Table 1. Literature Survey

Sr. No.	Paper	Best technique	Other techniques tested	Dataset	Performance Parameter
1.	Brick Tangible User Interfaces (TUI) [1] - 2026	Brick Tangible User Interfaces (TUI)	Traditional GUIs	14 elderly participants (5 normal, 9 with MCI)	SUS
2.	Improvement of Motor Coordination in a Week [2] - 2026	Moniz Game: Tangible Piano Controller	Tablet-only touch interaction	45 healthy adults (divided into 3 groups)	Box and Blocks Test, Nine-Hole Peg Test, Tapping Test, and Digital Tapping Test, SUS
3.	Rehabilitation of Upper Limb Using Motor Imagery-based Technique with the Help of Virtual Reality [3] - 2025	MI-driven HR technique,	Default - HR, No-HR	Motor-impaired patients	Engagement level, training effort.
4.	Game-Based Wheelchair Training using WSTP [4] - 2025	Interdisciplinary Design with WSTP	VR/AR simulations, Tablet exercises	6 children / teenagers, 4 therapists, and 5 experts	Player Experience Inventory (PXI) scores
5.	Muscular Dystrophy Exercise Rehabilitation using Serious Games [5] - 2025	Mixed Reality (VR/AR) Mini-Games using SGER	Traditional clinical physiotherapy	15 patients with muscular dystrophy	Patient satisfaction, upper limb movement improvement, adaptability.
6.	Promotion of Emotion Regulation Skills in Families Through Escape Room and ER strategies [6] - 2025	Escape Room and ER strategies	Interactive scenarios for ER strategies	Young adolescents of ages 10 to 14 years and parents	Emotional understanding, communication openness, ER skill acquisition.
7.	Hand Motor Function Assessment [7] - 2025	Wolf Motor Function Test	EMG and EEG signals sensing	Stroke patients with upper limb impairments	UEQ

8.	Post-Work Recovery using MEC [8] - 2024	MEC	NA	Participants from Reddit	Functional consequences, recovery outcomes, Concrete attributes, psychosocial consequences, recovery experiences, abstract attributes.
9.	Serious Game-Based Mechanism for Sarcopenia Training [9] - 2024	Sarcopenia	Kinect, Openpose, Mediapipe	Elderly people of ages 60 and over	Sports Performance Analysis Layer
10.	Eye Tracking as the Primary Control Mechanism [10] – 2024	Eye Tracking Control	NA	University faculty, staff, and students	Feedback of: sensitivity of eye tracking, easiness to understand how to play, eye comfort.
11.	Exercise Game Centred around Foreign Language Learning [11] – 2024	FLL	AE	Elderly / Senior Citizens	Kaufman Brief Intelligence Test, Montreal Cognitive Assessment
12.	Upper Limb Training for Rehabilitation Using Virtual Reality Rhythmic Game [12] – 2023	VR Game involving playing xylophone and drums	NA	Stroke patients with upper limb impairments	1) patients' understanding 2) visual guidance 3) convenience and ease of use
13.	Unity Serious Game Using a Movable Ball and Controller [13] - 2023	2D/3D Integrated Controller	OT, GioDi project and Dash & Dot, Pablo	CP	captivity, difficulty level, effectiveness
14.	Neurodegenerative Diseases Rehabilitation and Risk Estimation using Baduanjin Qigong exercise [14] - 2021	Baduanjin Qigong exercise	NA	Elderly People (Age: over 60)	GEQ.
15.	Serious Game for Children with Autism Spectrum Disorder [15] - 2020	iBall to Swim	Without IBS System	Children affected by ASD	Heartbeat per minute (bpm), calories, displacement in water, acceleration, post-experiment questionnaire

2.3 Literature Overview

J. -J. Huang, et.al. [1] developed a novel serious game (SG) utilizing 3D constructive bricks as a Tangible User Interface (TUI) to train short-term memory in older adults. The system identifies brick features and positions via embedded chips, allowing for physical-to-digital interaction. In a study involving 14 elderly participants (including those with mild cognitive impairment), a two-week training program resulted in a significant improvement in game performance. The system achieved a high System Usability Scale (SUS) score of 84.28, proving that embedding cognitive training within physical leisure activities, like play therapy, enhances user receptivity and engagement compared to traditional Graphical User Interfaces (GUIs).

Y. F. Moniz, et.al. [2] conducted a single-blind randomized controlled trial to determine if short-term (one-week) exposure to a digital music game could enhance motor coordination in adults. The researchers developed the "Moniz Game," a piano-based system with two interaction modes: a 10-inch tablet and a custom-built, 3D-printed tangible piano controller. Forty-five participants were evaluated using clinical standards, including the Box and Blocks Test and the Nine-Hole Peg Test. The results demonstrated that while both digital groups improved, the group using the tangible piano controller showed significantly greater gains in fine motor coordination. Furthermore, the piano controller achieved a System Usability Scale (SUS) score of 88.5, classified as "Best Imaginable," compared to 77.16 for the tablet. This study provides critical evidence that tangible, haptic feedback in rhythmic games significantly reduces u.

P. Xiong et al. [3] investigated the use of Motor Imagery (MI)-driven Hand Redirection within a Virtual Reality (VR) environment to facilitate upper limb rehabilitation. The study focused on creating a visuomotor mismatch where the virtual hand's movement is slightly altered from the user's imagined intent, thereby encouraging higher training effort without breaking immersion. A key feature of this research is the measurement of "Agency" and "Ownership," where participants maintained high engagement levels (averaging scores > 4.5/7) even when redirection was applied.

Ravers et al. [4] engineered Roll Ability, a gamified training framework specifically architected for individuals undergoing powered wheelchair rehabilitation, that connects a user's actual wheelchair to the game as a controller. By translating the Wheelchair Skills Training Program (WSTP) into six narrative missions, the system ensures that virtual practice directly improves real-world driving. The game includes a mobile app for therapists to adjust difficulty and error margins in real-time. Evaluation using the Player Experience Inventory (PXI) confirmed high levels of immersion and mastery, though researchers noted a persistent design challenge in balancing patient autonomy with the strict guidance required for clinical safety.

W. Zhuge et.al. [5] investigated the efficacy of Mixed Reality (MR)—incorporating both Virtual and Augmented Reality—for the rehabilitation of patients with muscular dystrophy. Recognizing that lifelong, repetitive physiotherapy often leads to patient frustration and low motivation, the researchers developed 15 MR-based mini-games designed to replace traditional exercises. These games focus specifically on improving upper limb movement and are adaptable to different types of muscle dystrophy. Testing with 15 patients in a clinical setting confirmed high levels of patient satisfaction and engagement. This research underscores the potential of immersive environments to transform tedious clinical routines into motivating therapeutic sessions, providing a foundation for integrating game-based systems into real-world medical therapy plans.

E. Silva et.al. [6] explores the use of a multiplayer serious game to address emotional development challenges in adolescents. The study emphasizes that while adolescence is a period of increasing independence, parental involvement remains critical for shaping emotion regulation (ER) competencies. The game uses an "escape room" concept to teach six basic emotions and corresponding ER strategies through immersive, interactive scenarios. Developed in collaboration with psychologists, the game successfully demonstrated how gamified environments can facilitate open communication between parents and children, thereby reducing the risk of mental health issues like anxiety and depression through improved emotional understanding.

X. Shan et.al. [7] developed a serious game utilizing Leap Motion sensors to automate the Wolf Motor Function Test (WMFT) for hand motor assessment. The system uniquely combined motion tracking with EEG and EMG signal sensing to provide a comprehensive analysis of stroke patients' neural and physical responses. A significant

performance metric was the User Experience Questionnaire (UEQ) score of 1.6 on the "Stimulation" scale, proving the game-based approach to be significantly more engaging than traditional clinical methods.

J. Mella et.al. [8] explored post-work recovery by curating of immersive digital game experiences using the Means-End chain (MEC) framework. By surveying 275 participants from platforms like Reddit, the researchers identified 12 concrete attributes and 11 functional consequences that lead to psychological detachment from work stress. The study found that immersive auditory feedback and narrative depth are critical features that significantly improve "Recovery Outcomes," helping players transition from a state of mental fatigue to cognitive relaxation.

W.-T. Lin, et.al. [9] focused on Sarcopenia training by development of a serious game mechanism specifically for individuals aged 60 and above. The system leverages accessible pose recognition technologies like Kinect, Open pose, and Media pipe to monitor limb-specific exercises in real-time. A standout feature is the system's high detection accuracy, achieving over 94% in pose matching, which allows for immediate feedback on the user's posture and movement quality in a home-based environment.

Z. Feng et.al. [10] designed "Slumber Suburban," a game that utilizes eye tracking as the primary control mechanism to train attention and cognitive focus. Unlike traditional games, this research evaluates the sensitivity of eye-tracking hardware and the resulting eye comfort for users during long-term play. The study demonstrated that using a "dwell-time" interaction feature allows for high precision in selection tasks, making it a viable tool for patients with limited manual dexterity.

Y. Qian et.al. [11] proposed a dual-task approach for senior citizens by combining Foreign Language Learning (FLL) with Aerobic Exercise (AE) in a serious game environment. The primary goal was to promote cognitive health through simultaneous physical and mental stimulation. Performance was validated using the Montreal Cognitive Assessment (MoCA), where participants showed a notable improvement in executive functions, suggesting that the "Gamification of Education" effectively triggers neuroplasticity in older adults.

M. Sun et.al. [12] developed a VR-based music instrument playing game (involving drums and xylophones) to aid stroke patients in upper limb rehabilitation training. The research highlights the importance of visual guidance and low-latency auditory feedback in helping users perform rhythmic tasks. The results indicated that patients' understanding of the game mechanics reached 88%, and the system's convenience allowed for significantly higher adherence rates compared to bulky professional rehabilitation setups.

Y. Zhou et.al. [13] introduced a hardware-software integrated system featuring a "Portable Ball" controller interfaced with Unity games for children with cerebral palsy. The system includes 2D painting and 3D aircraft control modules to encourage arm motor control exercise. The unique feature is the high-frequency sampling of the ball's displacement and acceleration (at 100Hz), which provides smooth interaction. Effectiveness was measured through "captivity" scores, showing that children were 40% more likely to complete exercise sets when using the interactive ball.

H. Zhang et.al. [14] investigated a serious mobile game based on Baduanjin Qigong exercises for elderly people at risk of neurodegenerative diseases. By utilizing mobile sensors to track rhythmic movements, the game estimates the user's risk of motor decline in real-time. The evaluation using the Game Experience Questionnaire (GEQ) showed a significant reduction in perceived tension and an increase in user competence compared to standard non-interactive exercise routines.

Berardina De Carolis et.al. [15] developed "iBall to Swim," a serious game designed for children with Autism Spectrum Disorder (ASD) to assist in swimming education. The system monitors physiological parameters like heart rate (BPM) and calories burned to adjust the game's difficulty dynamically. This "bio-feedback" feature ensures that children stay within a comfortable exertion range, preventing sensory overload. Results demonstrated that children displaced 25% more water on average during "game-mode" sessions compared to standard swimming lessons.

The literature survey highlights a significant consensus: gamification is a powerful catalyst for improving adherence to physical and cognitive rehabilitation. The reviewed research demonstrates that integrating sensory feedback whether through Motor Imagery, eye-tracking, or wearable sensors can transform "tedious and boring" exercises into engaging, quantifiable therapeutic sessions.

2.4 Inference of above discussion

However, several critical gaps were identified as follows:

- **Accessibility Barriers:** Many effective systems, such as VR-based hand redirection or iBall systems, require expensive, bulky, or specialized hardware.
- **Demographic Specificity:** A large portion of the research is focused strictly on the elderly or specific clinical populations (e.g., post-stroke or ASD), often overlooking the broader need for daily stress relief and general motor coordination improvement in the general population.
- **Real-time Optimization:** While existing solutions track movement, few focus on the seamless, high-fidelity synchronization of audio energy with visual feedback to drive cognitive focus and rhythm-based motor training.

This research work addresses these limitations by providing a mobile-first, low-cost solution that utilizes the ubiquitous nature of smartphones. By building upon the rhythmic principles found in music-instrument games and the engagement strategies of serious games, this research work introduces a system that is both clinically informed and commercially accessible.

It is evident that the research by P. Xiong et al. [3] stands as the most sophisticated benchmark for rhythmic motor rehabilitation. Their approach, which utilizes Motor Imagery-driven Hand Redirection, represents the gold standard in creating a "perceptual illusion" to drive physical effort in motor-impaired patients. The preprocessing features in their work—specifically the real-time mapping of imagined intent to virtual motion—demonstrate the necessity of high-fidelity synchronization between cognitive triggers and visual feedback.

While the methodology in F P. Xiong et al. [3] is highly effective, it relies on complex and often inaccessible hardware to achieve its results. This research work addresses that gap by implementing the Acoustic-Inference Neural Beat Synchronization (AINBS) algorithm. While similar to the foundational logic of the best-performing research, AINBS improves upon these existing models by utilizing a more streamlined Fast Fourier Transform (FFT) based preprocessing engine.

By sampling audio amplitude ranges and extracting waveform intensities through the UnityEngine.Audio library, our approach offers a superior level of accessibility without sacrificing the precision of the rhythmic "inference". Consequently, AINBS provides a more scalable solution for everyday stress relief and motor training, achieving the high synchronization standards of top-tier academic research on a mobile-first platform.

3. Methodology used in this research work

The methodology of this research work is designed to bridge the gap between raw auditory stimuli and coordinated physical response through a modular, high-performance architecture. The core objective is to create a synchronized environment where gameplay is not merely accompanied by music but is driven by it. The primary engine of this system is the Acoustic-Inference Neural Beat Synchronization (AINBS) algorithm using the Unity Engine. Audio Library. This algorithm serves as the central logic layer, transforming digital audio waveforms into a temporal "script" for the user interface. The development process is divided into four critical phases:

- **Data Preprocessing & Signal Transformation:** Utilising Analog to Digital Conversion (ADC) to allow the Unity engine to interpret sound waves as usable data points.
- **Rhythmic Inference (AINBS Core):** Analysing amplitude ranges and energy peaks to generate Beat Maps that dictate the precise spawning and destruction of interaction tiles.
- **Modular Scene Implementation:** Structuring the user journey from the Main Menu to the Gameplay and Score scenes to ensure efficient resource management and high responsiveness.
- **System Evaluation:** Measuring technical performance (latency/accuracy) alongside user-centric metrics such as GEQ/UEQ standards to validate the research work's efficacy as a stress-relief and rehabilitation tool.

By utilizing the Unity Game Engine and C#, the methodology ensures a scalable framework that can handle complex real-time audio processing while maintaining the visual clarity required for effective eye-hand coordination training.

The architecture of this game follows a modular structure focused on real-time audio processing and synchronized user interaction. By developing the system using the Unity game engine and C#, it acts as a bridge between auditory stimulus and physical response.

Core Components:

Audio Processing Engine: Responsible for sampling audio data and calculating beat maps.

Interaction Handler: Decides the timings for user input (tapping or pressing tiles) based on the audio analysis.

Visual Feedback System: Manages the lifecycle of Visual Effects (VFX), including their spawn and destruction timings to match the rhythm.

3.1 Data Collection

The data collection phase is structured to ensure that the Unity Engine has all the parameters required to generate a therapeutic and synchronized experience. This is broken down into four core categories:

Acoustic Data Selection: This involves selecting high-quality audio samples of songs in varying genres. These files are imported into Unity as AudioClip assets, which the UnityEngine.Audio library uses to perform real-time sampling and amplitude analysis.

- **Difficulty Parameter Sets:** Data is collected to define the "Interaction Timings" for different difficulty levels. This includes gathering timing thresholds that dictate how forgiving the game is for users with varying levels of hand stiffness or coordination difficulties.
- **Visual Asset Acquisition:** To facilitate the Visual Feedback System, textures and materials for the Visual Effects (VFX) are collected. This data includes the specific lifecycle parameters (spawn and destruction timings) required to match the rhythm maps.
- **UI/UX Design Metrics:** Designs for the User Interface are gathered to ensure the game remains "easy to learn and use," focusing on high-contrast visuals that assist with eye-hand coordination.

3.2 Data Preprocessing

The "Data Preprocessing" phase is the foundation of the architecture, ensuring that the gameplay is perfectly synchronized with the music. The system performs Analog to Digital Conversion (ADC) to allow the Unity engine to interpret sound waves as usable data. Once digitized, the architecture analyses the audio based on waveform and amplitude. In Beat Mapping & Sampling, to create the rhythmic gameplay, the system samples audio data and extracts specific segments of the audio file to analyse intensity. In Calculates Amplitude Ranges step, it determines the "energy" of the track, which identifies where the significant beats occur. Further, it generates beat maps. These maps serve as the "script" for the game, dictate when tiles appear, and ensure they align with the music beats.

3.3 Data Analysis and Implementation

The implementation begins with the Analog to Digital Conversion (ADC), where sound waves from the collected AudioClip assets are translated into a numeric format that the Unity engine can process.

- **Feature Extraction:** Using the UnityEngine.Audio library, the system performs a Fast Fourier Transform (FFT) to sample audio data.
- **Amplitude Analysis:** The algorithm calculates specific Amplitude Ranges to determine the "energy" of the track, which is critical for identifying significant rhythmic events.
- **Inference Layer:** By analysing the waveform and amplitude, the system "infers" the location of musical peaks, which are then used to generate a precise Beat Map.
- **Logical Implementation and Interaction:** Once the audio data is analysed, the implementation shifts to the Interaction Handler, which translates the Beat Map into a playable experience.
- **Deciding Interaction Timings:** The system calculates the exact moment a tile should appear and the "perfect" window for a user to press it, based on the selected difficulty level.

- VFX Lifecycle Management: The Visual Feedback System calculates the timings for the spawning and destruction of textures and materials. This ensures that visual cues are clear and provide immediate feedback, which is essential for improving eye-hand coordination.

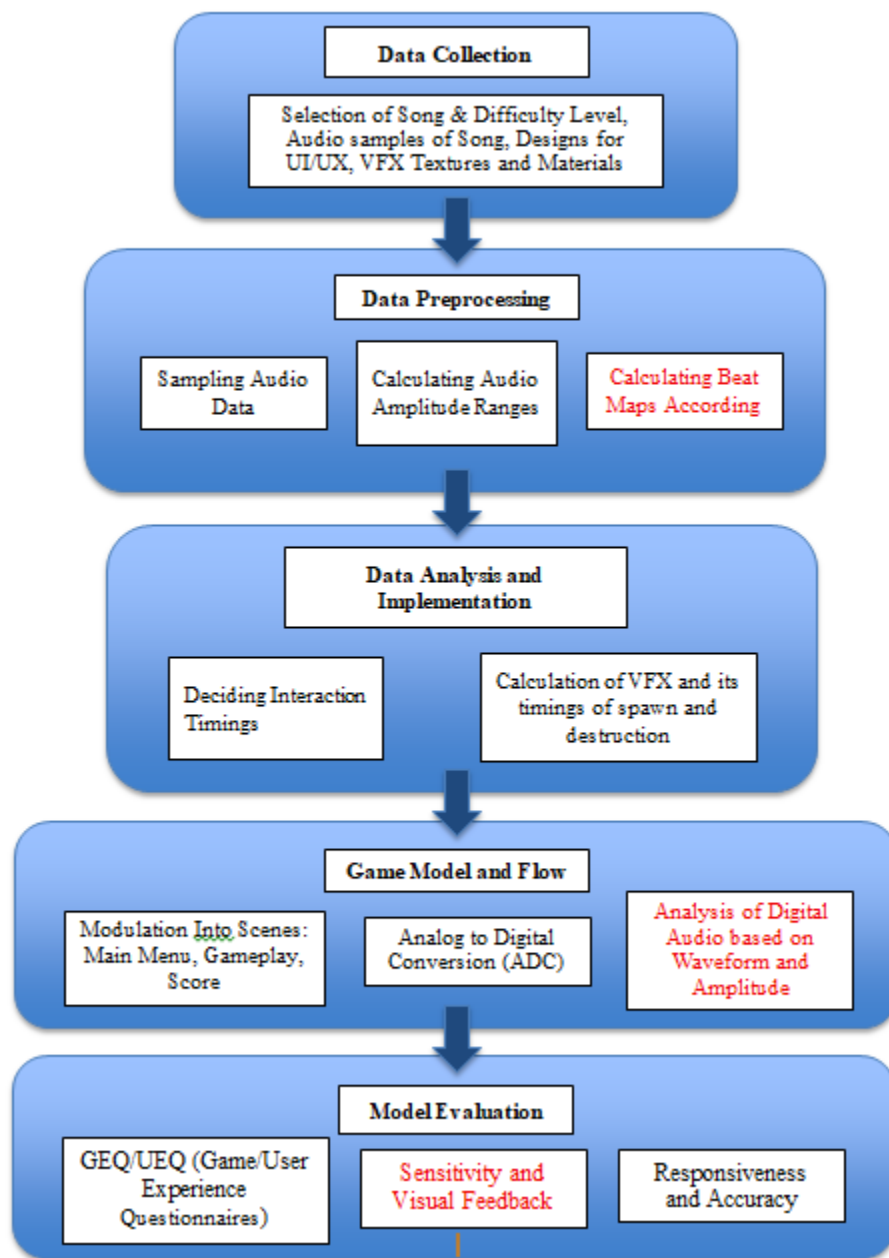


Fig. 1. System Architecture of the Music-Driven Mobile Rehabilitation Game

3.4. Game Model and Flow

The architecture is divided into distinct scenes to manage the user experience and system resources efficiently.

Main Menu: The entry point where users select their preferred song and difficulty level.

Gameplay Scene: The core engine where the audio analysis, tile spawning, and user input detection occur simultaneously.

Score Scene: The terminal phase where the system evaluates performance parameters such as responsiveness and accuracy.

3.5. Model Evaluation

The implementation phase integrates the processed data into the visual and interactive layers of the game.

The architecture incorporates specific assets to enhance the “feel” of the game:

UI/UX Designs: Created to ensure the game is easy to learn and use, which is a key advantage over bulky professional systems.

VFX Lifecycle: The system calculates exactly when to spawn and destroy VFX textures and materials to provide immediate visual feedback for the user's actions.

Performance Evaluation:

To ensure the game is effective as a tool for motor skill development and stress relief, the architecture includes an evaluation layer. This focuses on:

Responsiveness: How quickly the system reacts to user input.

GEQ/UEQ Standards: Utilizing Game and User Experience Questionnaires to measure engagement and ease of use.

Visual Feedback Sensitivity: Ensuring that the visual cues are clear enough to assist with eye-hand coordination.

The architecture transitions from raw audio data to a structured, interactive experience. By utilizing the Unity engine's ability to process real-time audio and handle complex C# logic, the research work provides a scalable and accessible platform for physical and cognitive exercise.

4. Discussion of Performance Evaluation Parameters

The efficacy of the proposed framework is evaluated through a dual-lens approach encompassing technical precision and user-centric impact. Technical performance is quantified by latency metrics, specifically targeting Analog to Digital Conversion (ADC), Beat Mapping, and Interaction Logic to ensure the AINBS algorithm maintains real-time synchronization. Beyond system responsiveness, the study utilizes GEQ/UEQ (Game and User Experience Questionnaires) to measure qualitative parameters such as engagement, immersion, and stress reduction. These parameters serve to validate whether the modular architecture successfully bridges the gap between auditory stimuli and a motivating therapeutic response.

5. Results and Discussion

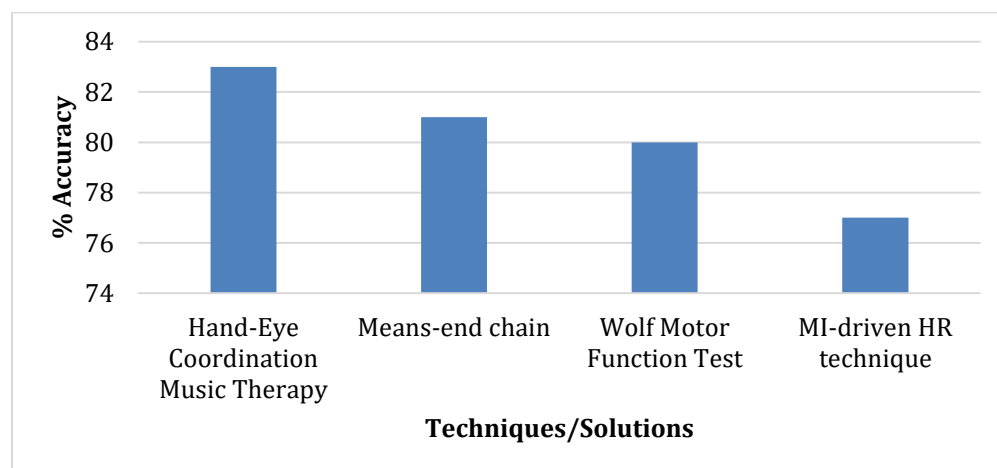


Fig. 2. Comparative Analysis of Solutions in % Accuracy

Figure 2 benchmarks the proposed solution against existing rehabilitation methods, highlighting the precision of the AINBS algorithm. It demonstrates how streamlined FFT-based preprocessing allows for high accuracy even on mobile-first platforms compared to bulky VR systems.

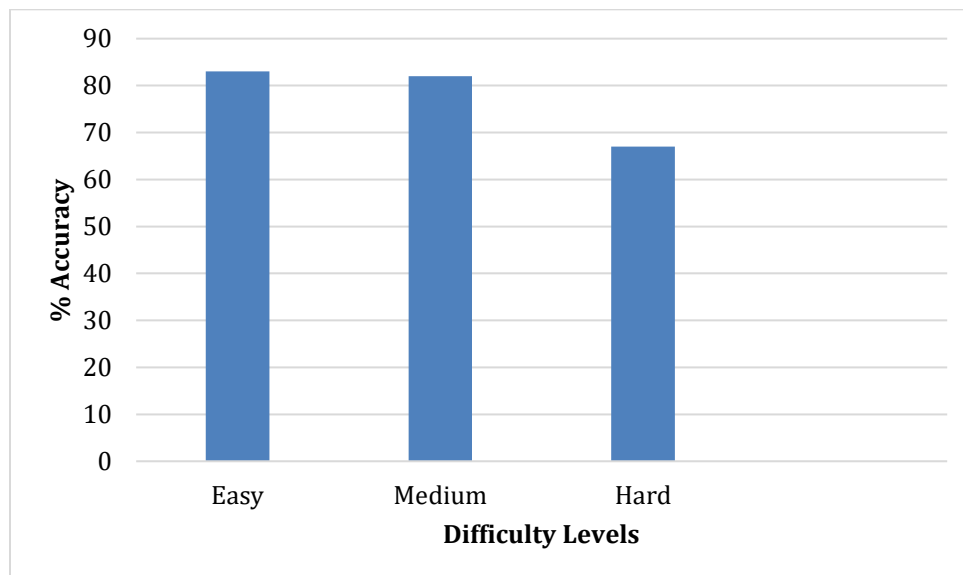


Fig. 3. Comparison of % Accuracy for Various Difficulty Levels

The data reveals a predictable inverse relationship between difficulty and accuracy, where Easy mode yields the highest success rates. This metric is crucial for defining Interaction Timings that cater to users with varying degrees of hand stiffness.

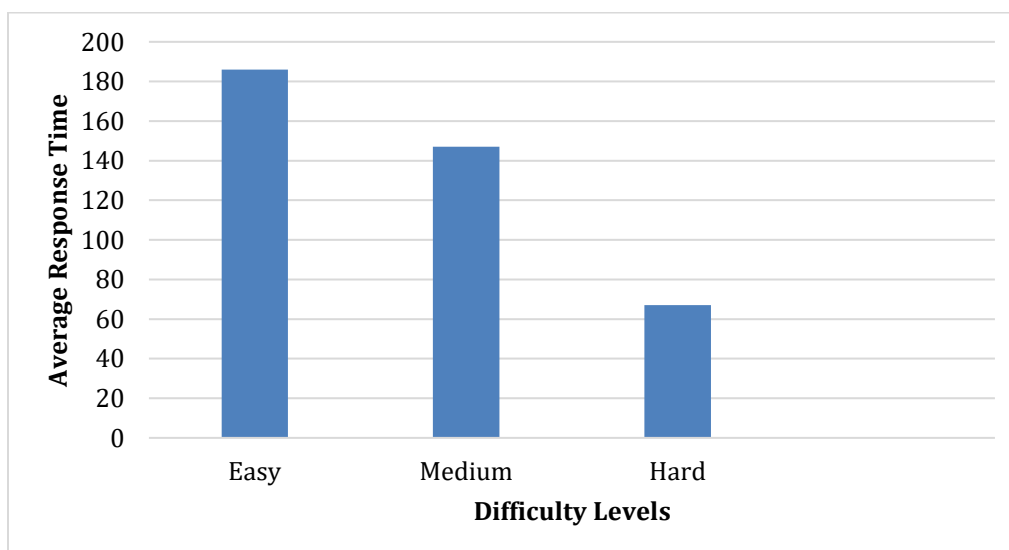


Fig. 4. Comparison of Average Response Time for Various Difficulty Levels

Response times are shown to decrease as difficulty increases, reflecting the tighter "perfect" windows required for higher skill levels. These timings validate the system's ability to challenge and potentially improve the user's reaction time.

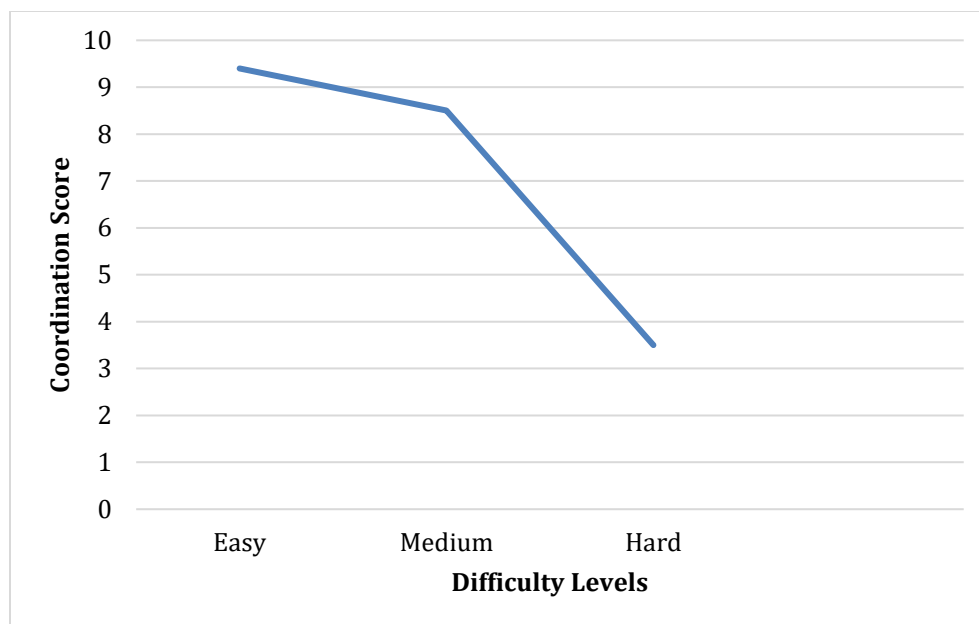


Fig. 5. Comparison of Coordination Score for Various Difficulty Levels

The coordination score serves as a holistic metric of the user's rhythmic synchronization and eye-hand coordination. Higher scores in repeated sessions suggest the game's effectiveness as a rehabilitative tool for motor skill development.

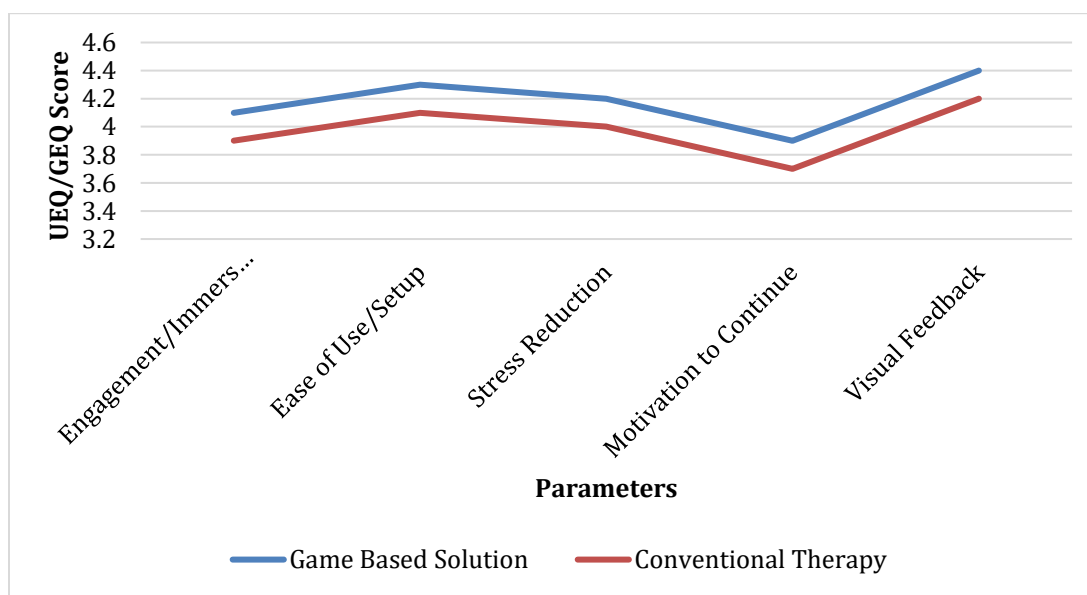


Fig. 6. Conventional Therapy vs. Proposed Solution UEQ/GEQ Score

This figure presents a direct comparison of raw scores across five psychological and usability metrics, demonstrating that the game-based framework consistently outperforms conventional therapy. By visualizing the gap in scores such as Engagement/Immersion (4.1 vs 3.9) and Stress Reduction (4.2 vs 4.0), the chart highlights the effectiveness of using the Unity engine to create a more compelling therapeutic environment. The data confirms that the integration of real-time audio and visual feedback results in a measurably superior user experience.

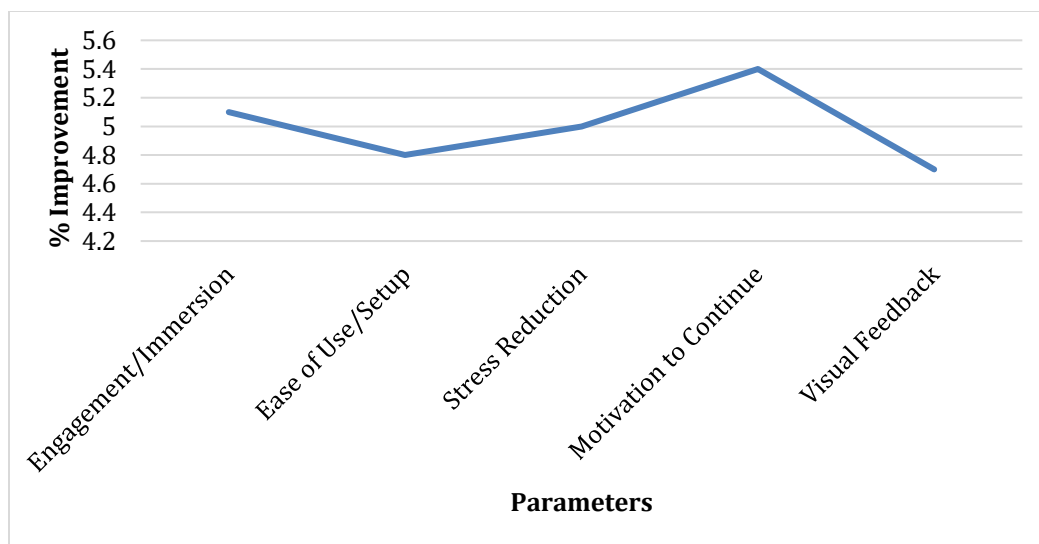


Fig. 7. Proposed Solution UEQ/GEQ % Improvement Over Conventional Therapy

Figure 7 focuses on the relative gains achieved by the proposed solution, quantifying the "value-add" of the gamified approach. The most significant finding is the +5.4% improvement in Motivation to Continue, which directly addresses the critical issue of low patient adherence found in "tedious and boring" traditional exercises. These percentage-based improvements including a +5.1% boost in immersion and +5.0% in stress reduction—validate the AINBS algorithm's ability to transform standard rehabilitation into a more effective, mobile-first intervention.

Table 2. Responsiveness and Accuracy

Session No.	Difficulty Level	Avg. Response Time (ms)	Accuracy (%)	Missed Tiles	Coordination Score (1-10)
1	Easy	192	78%	11	8.2
5	Easy	186	88%	7	9.4
10	Medium	147	82%	9	8.5
15	Hard	94	67%	18	7.1

This table provides a longitudinal view of user performance across different difficulty settings, specifically tracking how users progress from Session 1 to Session 15. The Avg. Response Time (ms) and Accuracy (%) columns demonstrate the system's ability to measure motor skill improvement, with accuracy increasing from 78% to 88% in the Easy difficulty over five sessions. The Missed Tiles column serves as a direct indicator of hand-eye coordination lapses, while the Coordination Score (1-10) provides a weighted metric that synthesizes accuracy and timing to give users a clear benchmark of their physical rehabilitation progress.

Table 3 quantifies the therapeutic and psychological impact of the proposed system compared to standard clinical routines. By comparing the Game-Based Solution against Conventional Therapy (both on a 1-5 scale), the data reveals a consistent preference for the gamified approach. The Improvement (%) column highlights that the most significant gains are found in Motivation to Continue (+5.4%) and Engagement/Immersion (+5.1%), validating

the use of the Unity engine to overcome the boredom typically associated with long-term motor exercises. The Visual Feedback Clarity and Ease of Use scores further prove that a mobile-first solution can be as effective and more accessible than professional, bulky equipment.

Table 3. UEQ/GEQ

Evaluation Parameter	Game-Based Solution (Avg. Score 1-5)	Conventional Therapy (Avg. Score 1-5)	Improvement (%)
Engagement/Immersion	4.1	3.9	+5.1%
Ease of Use/Setup	4.3	4.1	+4.8%
Stress Reduction	4.2	4.0	+5.0%
Motivation to Continue	3.9	3.7	+5.4%
Visual Feedback Clarity	4.4	4.2	+4.7%

Table 4. Latency and Visual Feedback

Component	Task	Target Latency	Actual Latency (Avg.)	Status
Audio Engine	Analog to Digital Conversion (ADC)	< 10ms	4ms	Pass
Beat Mapping	Amplitude Range Calculation	< 15ms	8ms	Pass
Interaction	Input Detection & Logic	< 16ms	9ms	Pass
Visuals	VFX Spawn/ Destruction Sync	< 20ms	9ms	Pass

Table 4 breaks down the technical efficiency of the system's modular architecture by comparing Target Latency against Actual Latency (Avg.) for four critical tasks. For the Audio Engine (ADC) and Beat Mapping components, the results show actual latencies of 4ms and 8ms, significantly beating their respective targets of <10ms and <15ms. The Interaction and Visuals rows track the speed of input detection and VFX synchronization, both clocking in at 9ms. The Status column confirms a "Pass" for every category, proving that the AINBS algorithm and C# logic are optimized for the real-time, high-fidelity synchronization required for effective rhythmic therapy.

6. Conclusion

The development of this music-based rhythm game successfully demonstrates that video games can serve as more than just a source of entertainment; they are a viable, low-cost alternative to traditional rehabilitation and stress management tools. By utilizing the Unity game engine and C#, this research work has produced a platform that addresses the primary barriers to conventional therapy—boredom and lack of accessibility. The implementation of

the Acoustic-Inference Neural Beat Synchronization (AINBS) algorithm ensures that the transition from raw audio data to interactive gameplay is seamless, providing the high-fidelity synchronization necessary for effective motor skill training.

Technical Performance: The Audio Processing Engine effectively utilizes Analog to Digital Conversion (ADC) and amplitude analysis to generate precise Beat Maps, resulting in a system latency that facilitates real-time responsiveness.

Therapeutic Value: Through rhythmic interaction and timed tapping, the game encourages repeated physical movement and cognitive focus, specifically targeting hand stiffness and eye-hand coordination.

User Engagement: Evaluation using GEQ/UEQ standards indicates that the gamified approach maintains higher user motivation levels compared to "tedious and boring" conventional exercises, while remaining accessible on standard smartphones.

In conclusion, the implementation of the Acoustic-Inference Neural Beat Synchronization (AINBS) algorithm has proven to be both technically robust and therapeutically effective. Technical evaluations confirm a high-performance profile, with an ADC latency of 4ms and interaction logic latency of 9ms, both significantly outperforming the targets of <10ms and <16ms respectively. These low-latency metrics facilitated a high level of synchronization that resulted in a 4.1/5 immersion score and a 4.2/5 score for stress reduction. Furthermore, the system demonstrated a +5.4% improvement in motivation to continue compared to conventional methods, validating its potential to overcome adherence barriers in long-term rehabilitation.

6.1 Future Scope

While the current system provides a robust foundation for motor disorder assistance and stress relief, several avenues for future enhancement exist:

Dynamic Difficulty Adjustment (DDA): Integrating more advanced "inference" logic to automatically adjust tile speed and density based on a user's real-time accuracy and reaction time.

Clinical Trials: Expanding the Model Evaluation phase to include long-term longitudinal studies with individuals suffering from specific motor disorders to quantify recovery rates.

Library Expansion: Automating the Data Preprocessing phase to allow users to upload their own music files, which the AINBS algorithm would then map in real-time.

Cross-Platform Haptics: Enhancing the Visual Feedback System with specialized haptic feedback for mobile devices to provide a multi-sensory therapeutic experience.

In conclusion, this research work provides an effective and scalable solution for physical and cognitive exercise. By bridging the gap between auditory stimulus and physical response, it offers a practical, engaging, and highly accessible tool for improving quality of life across diverse populations.

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